Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	1200	**2**	**M1** for 1962 ÷ 1.635
1(a)(ii)	1667.7[0] final answer	**2**	**M1** for $1962 \times (1 - \frac{15}{100})$ oe or **B1** for 294.3[0] If 0 scored, **SC1** for answer 1020
1(a)(iii)	275	**2**	**M1** for 220 ÷ *their* (5 – 1) soi
1b(i)	165	**3**	**M2** for $\frac{9752-3680}{3680}[\times 100]$ oe or $\frac{9752}{3680}\times 100$ oe or **M1** for $\frac{9752}{3680}$ or 9752 – 3680
1b(ii)	51 200	**3**	**M2** for $\frac{74240}{100+45}[\times 100]$ oe or **M1** for 74 240 associated with 145[%] oe
2(a)	–1.5	**3**	**M1** for $30 + 2x = 9 - 12x$ or $10 + \frac{2}{3}x = 3 - 4x$ **M1** for collecting *their* terms correctly to reach $ax = b$
2(b)	$6ab^2(2b + 3a^2)$ final answer	**2**	**M1** for any correct partial factorisation seen or for correct answer seen
2(c)(i)	$10a^5c^9$ final answer	**2**	**B1** for final answer with $10a^kc^9$ or $10a^5c^k$ or ka^5c^9
2(c)(ii)	$\frac{8a^6}{c^9}$ or $8a^6c^{-9}$ final answer	**2**	**B1** for final answer with $\frac{8a^6}{c^k}$ or $\frac{8a^k}{c^9}$ or $\frac{ka^6}{c^9}$ $[k \neq 0]$ or for correct answer seen

Question	Answer	Marks	Partial Marks
2(d)	0.5 or $\frac{1}{2}$	**3**	**M1** for $y = \frac{k}{(x+2)^2}$ oe **B1** for $k = 50$ or **M2** for $2(3 + 2)^2 = y(8 + 2)^2$ oe
2(e)	$\frac{7x-x^2}{2(x-2)}$ or $\frac{7x-x^2}{2x-4}$ oe final answer	**3**	**M1** for $5 \times 2 - (x - 5)(x - 2)$ oe seen **M1** for common denominator $2(x - 2)$ oe isw
3(a)	Rotation 90[°] clockwise oe Origin oe	**3**	**B1** for each
3(b)(i)	Image at (–4, –1) (–4, –4) (–2, –4)	**1**	
3(b)(ii)	Image at (3, –1) (5, –1) (3, –4)	**2**	**B1** for translation by $\begin{pmatrix} 7 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -5 \end{pmatrix}$ or for 3 correct points not joined
3(b)(iii)	Image at (–2, ½) (–2, 2) (–1, 2)	**3**	**B2** for 3 correct co-ordinates soi in working or correct size and orientation in wrong position or **M1** for $\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}\begin{pmatrix} -4 & -4 & -2 \\ 1 & 4 & 4 \end{pmatrix}$ shown or for statement: enlargement, sf 0.5, (0, 0)
4(a)	$\frac{1}{2} \times 4(x - 1) \times (2x + 5)[\sin 90] = 30$ oe	**M1**	
	$8x^2 - 8x + 20x - 20$ or better	**B1**	correct expansion of brackets
	Completion to $2x^2 + 3x - 20 = 0$	**A1**	with no errors or omissions seen
4(b)	$(2x - 5)(x + 4)$	**M2**	Allow **M2** for e.g. $2x(x + 4) - 5(x + 4)$ then $2x - 5[= 0]$ and $x + 4[= 0]$ **M1** for $2x(x + 4) - 5(x + 4)$ or $x(2x - 5) + 4(2x - 5)$ or $(2x + a)(x + b)$ [= 0] where $ab = -20$ or $a + 2b = 3$ [a, b integers]
	2.5 and –4 cao	**B1**	